## Normal family of analytic functions

Defn: Let $U$ be a nonempty open subset of $\mathbb{C}$, and let $g_k : U \to \mathbb{C}$, $k=1,2,\cdots$, be a family of complex analytic functions. The family $\{g_k\}$ is said to be <u>normal on $U$</u> if every sequence of functions selected from $\{g_k\}$ has a subsequence that converges uniformly on every compact subset of $U$, either to a bounded analytic function or to $\infty$.

The family $\{g_k\}$ is normal at a point $w \in \mathbb{C}$ if $\exists$ an open set $U$ containing $w$ such that $\{g_k\}$ is normal on $U$.



<u>Montel's theorem</u>:

Let $\{g_k\}$ be a family of complex analytic functions on an open domain $U$. If $\{g_k\}$ is not a normal family, then for all $w \in \mathbb{C}$ with at most one exception $g_k(z) = w$ for some $z \in U$ and for some $k$.

Using the Montel's theorem, we get the following characterization of Julia sets.

<u>Theorem</u>: $J(f) = \{ z \in \mathbb{C} : \text{the family } \{f^k\}_{k \in \mathbb{N}} \text{ is not normal at } z \}$.

<u>Proof</u>: Let $z \in J(f)$. Then $\{f^k(z)\}$ is bounded. Also, in any neighborhood $V$ of $z$, $\exists\, w \in V$ s.t. $f^k(w) \to \infty$.

$\Rightarrow$ $\{f^k\}$ is not normal on $V$.

Suppose $z \notin J(f)$.
Then either $z \in \operatorname{int} K(f)$ or $z \in \mathbb{C} \setminus K(f)$.

<u>If $z \in \operatorname{int} K(f)$</u>, then $\exists$ an open set $V$ s.t.
$z \in V \subseteq \operatorname{int} K(f)$

Then for all $w \in V$, $f^k(w) \in K(f)$ $\forall k \in \mathbb{N}$.
$\Rightarrow \{f^k\}$ is normal at $z$ (by Montel's thm.)

<u>If $z \in \mathbb{C} \setminus K(f)$</u>, $|f^k(z)| > r$ for some $k$,
where $r$ is as in the previous lemma.
$\Rightarrow |f^k(w)| > r$ $\forall w$ in some neighbourhood $V$ of $z$ (by continuity of $f^k$)
$\Rightarrow f^k \to \infty$ uniformly on $V$.
$\Rightarrow \{f^k\}$ is normal at $z$.

$\therefore$ If $\{f^k\}$ is not normal at $z$, $z \in J(f)$



<u>Lemma</u>: Let $f$ be a polynomial of degree $\geq 2$.
Let $w \in J(f)$ and let $U$ be any neighborhood of $w$.
Then, for each $j \in \mathbb{N}$, the set

$$W = \bigcup_{k=j}^{\infty} f^k(U)$$

is the whole of $\mathbb{C}$, except possibly for a single point. Any such exceptional point is not in $J(f)$, and is independent of $w$ and $U$.

<u>Proof</u>: Since $w \in J(f)$, by the previous theorem, the family $\{f^k\}_{k=j}^{\infty}$ is not normal at $w$.
Then Montel's theorem says that $W$ is the whole of $\mathbb{C}$, except possibly a single point.

Suppose $v \notin W$.
We have to show that $v \notin J(f)$.
For this we'll show that the family
$\{f^k\}$ is normal at $v$.

Let $f(z) = v$ for some $z \in \mathbb{C}$.
Then $z \notin W$ (because $f(W) \subseteq W$ & $v \notin W$)
Since, $\mathbb{C} \setminus W$ has at most one point,
$z = v$.
$\Rightarrow$ $f(z) - v$ is a polynomial of degree $n$ having $z = v$ as the only root.
$\Rightarrow$ $f(z) - v = c(z-v)^n$ for some constant $c$.
If $z$ is sufficiently close to $v$, then
$f^k(z) - v \to 0$ as $k \to \infty$,
the convergence is uniform on $\{z : |z-v| < (2|c|)^{\frac{-1}{n-1}}\}$



$\Rightarrow$ $\{f^k\}$ is normal at $v$.
$\Rightarrow$ $v \notin J(f)$.
Clearly, $v$ only depends on the polynomial $f$.

Corollary:
(a) The following holds for all $z \in \mathbb{C}$ with at most one exception: if $U$ is an open set that intersects $J(f)$ then $f^{-k}(z)$ intersects $U$ for infinitely many values of $k$.
If there is an exceptional point $z$, $z \notin J(f)$.

(b) If $z \in J(f)$, then $J(f)$ is the closure of
$\bigcup_{k=1}^{\infty} f^{-k}(z)$.

Proof:

(a) Let $z \in \mathbb{C}$ be such that it is not the exceptional point in the previous lemma.

Then $z \in f^k(U)$ for some $k \geq 1$.

$\Rightarrow f^{-k}(z) \cap U \neq \phi$.

(b) If $z \in J(f)$, then $f^{-k}(z) \subseteq J(f)$ $(\because f^{-1}(J) \subseteq J)$

$\Rightarrow \bigcup_{k=1}^{\infty} f^{-k}(z) \subseteq J(f)$

Since $J(f)$ is closed, $\overline{\bigcup_{k=1}^{\infty} f^{-k}(z)} \subseteq J(f)$

Also, if $z \in J(f)$ and $U$ is any open set containing $z$, then $f^{-k}(z) \cap U \neq \phi$ for some $k$ by (a).

$\therefore z \in \overline{\bigcup_{k=1}^{\infty} f^{-k}(z)}$.